
SEC Form 4 Data Pipeline

Results Report 2025

Module IT for Finance I

University TH Köln - Campus Südstadt

Group Group 01

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1. Overview

This report presents the results of our insider transaction analysis for the year 2025. The pipeline processed all four quarters of SEC Form 4 bulk data and produced monthly rankings of the top five insider purchases and bottom five insider sales across three metrics: shares traded, number of transactions, and USD nominal volume. What follows is a summary of the key patterns we found, how the three metrics compare, and what the data suggests from a financial perspective.

2. Year-Level Trends

Figure 1 shows total insider purchase and sale volume by month. Sale volume exceeds purchase volume in almost every month, which is consistent with what academic research predicts. Insiders sell for all kinds of reasons - liquidity needs, portfolio diversification, pre-planned 10b5-1 schedules - so high sale volume alone does not necessarily signal bearish sentiment. Purchases, on the other hand, are more deliberate. An insider buying stock with their own money usually means they think it's undervalued.

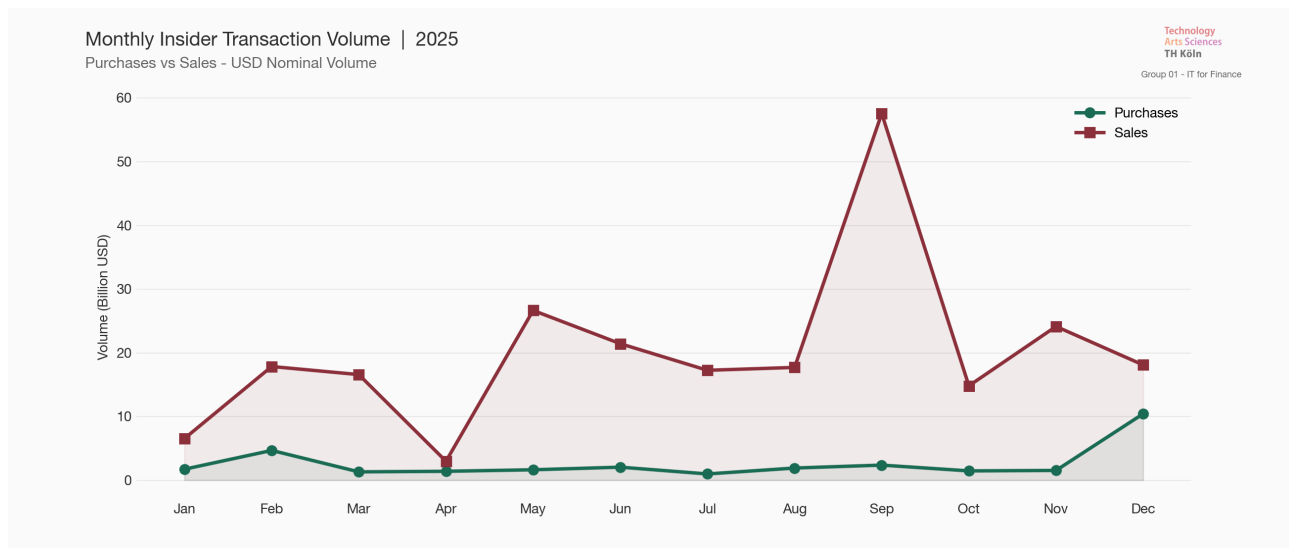


Figure 1: Monthly insider transaction volume (purchases vs. sales)

Figure 2 translates that into a purchase-to-sale ratio. A value above 1.0 would mean insiders collectively bought more than they sold. In 2025, no month crossed that line - sales dominated throughout. December came closest at 0.58, driven largely by a single \$7.6 billion purchase by Merus

N.V., but even that was not enough to tip the balance. March sat at just 0.08 - purchase activity barely registered that month (the top buyer came in at \$153 million) while sales topped \$3 billion.

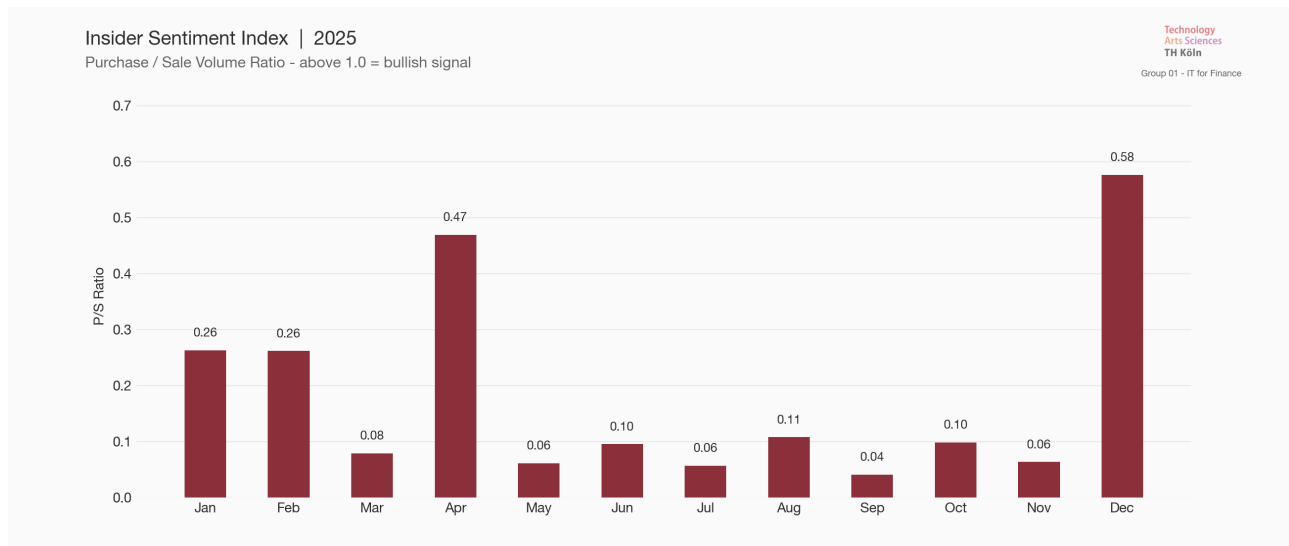


Figure 2: Insider sentiment index (purchase/sale volume ratio)

3. Recurring Sellers

One of the more interesting outputs is the repeat-seller heatmap (Figure 3). It filters for companies that appear in the Top-5 sales ranking across at least two different months. Oracle (ORCL) shows up in January and June, Dell (DELL) in January, June, and October, and CoreWeave (CRWV) across three consecutive months from August through October. Eli Lilly (LLY) also appears three times - in February, October, and November. For Oracle and Dell, executive compensation in the form of stock options likely explains the pattern - insiders exercise and sell as part of their regular schedule. CoreWeave's cluster is different: it went public in 2025 and the concentrated selling probably reflects lock-up expirations rather than a bearish signal. This kind of context matters - programmatic selling is far less informative than a sudden burst from executives who had not sold before.

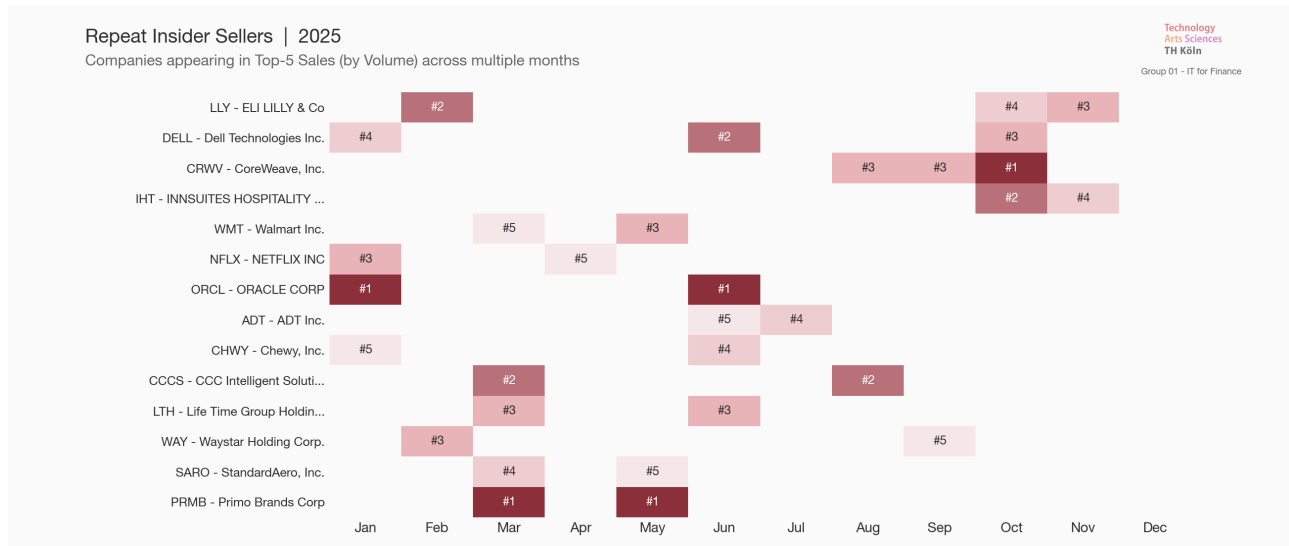


Figure 3: Companies appearing in Top-5 sales across multiple months

4. Metric Comparison

The three metrics we computed tell different stories about the same data, and comparing them is where the analysis gets interesting. Table 1 summarizes what each metric captures and where it falls short.

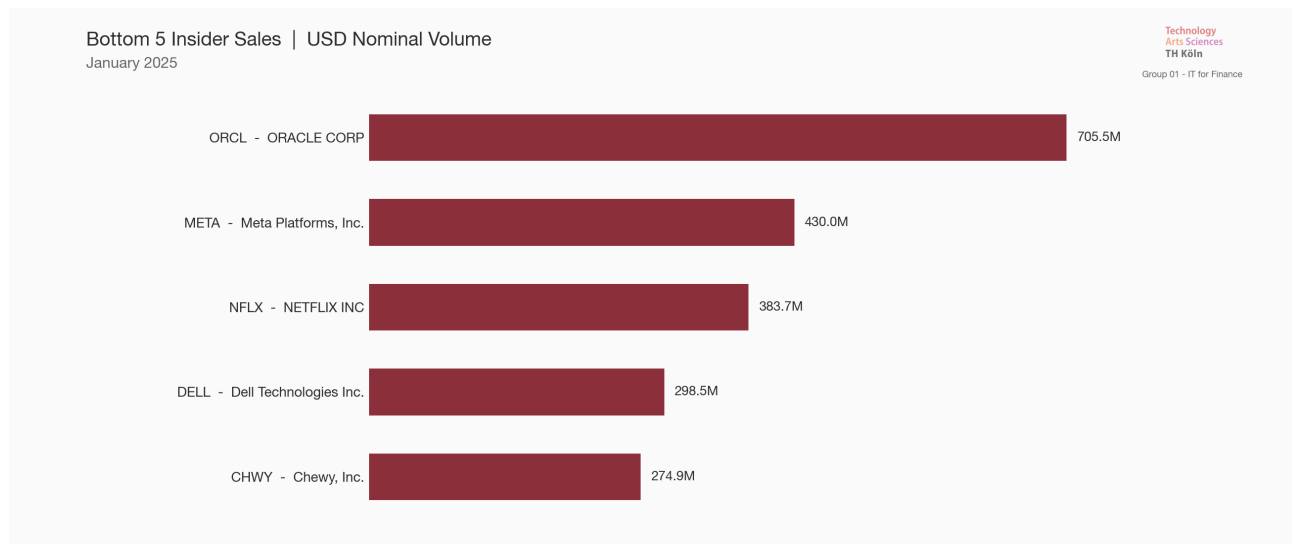
Table 1: Metric characteristics and limitations

Metric	What It Captures	Limitation
Shares Traded	Raw number of shares bought or sold	Misleading for low-price stocks (high count, low value)
Number of Transactions	How often insiders traded	Many small trades can outweigh one large meaningful trade
USD Nominal Volume	Shares multiplied by price per share	Missing prices reduce coverage (especially derivatives)

In practice, the differences are obvious. Table 2 shows the January 2025 Top-5 purchases ranked by USD volume. TKO Group Holdings leads at \$455 million. But the top purchaser by shares traded that same month was Norris Industries (NRIS) with 17.4 million shares - total value just \$1.4 million, a penny stock that doesn't even appear in Table 2. And by transaction count, RENN Fund (RCG) topped the list with 140 individual purchases totaling only \$64,000. Figures 4 and 5 contrast the purchase and sale side for the same month - completely different companies dominate each.

Table 2: Top 5 insider purchases by USD volume - January 2025

Company	Ticker	Shares	Transactions	USD Volume
TKO Group Holdings	TKO	3,013,743	120	\$454,960,681
Immunovant	IMVT	16,845,010	1	\$336,900,200
CION Ares Div. Credit	CADCX	6,040,000	5	\$151,000,000
Keurig Dr Pepper	KDP	3,619,600	1	\$120,677,464
JH CQS Asset Backed	—	3,750,000	2	\$75,000,000

*Figure 4: Top 5 insider purchases by shares traded - January 2025**Figure 5: Bottom 5 insider sales by USD nominal volume - January 2025*

September tells a similar story from a different angle: Tesla (TSLA) dominated the volume ranking at nearly \$1 billion in purchases across 25 transactions, but it did not even appear in the top five by share count because Tesla's high stock price means fewer shares per dollar spent. Each metric alone paints an incomplete picture. Companies that show up across all three rankings signal the strongest insider conviction.

5. Financial Interpretation

Lakonishok and Lee (2001) demonstrated that aggregate insider purchasing activity has predictive power for future stock returns. Our data supports this framing. The most informative signal is not any single executive buying shares, but rather clusters of buying activity at the same company from multiple insiders. When a CEO, CFO, and two directors all buy within the same month, that is hard to explain away as routine compensation planning.

On the sales side, the data is noisier. Most insider selling in our dataset follows predictable patterns - large blocks sold at regular intervals, often linked to vesting schedules or 10b5-1 plans. The repeat-seller heatmap confirms this: the same names recur month after month. That kind of systematic selling carries little informational value. The cases worth investigating are sudden, one-off sales by executives who had not sold in previous months - those are rarer in the data but would warrant further analysis beyond the scope of this pipeline.

Overall, USD nominal volume proved to be the most reliable metric for identifying economically meaningful insider activity, while shares traded and transaction count served as useful cross-checks to separate genuine conviction from noise.

In summary, the 2025 data paints a clear picture: insider selling dominated every month, but the nature of that selling was overwhelmingly routine. The purchase side, though smaller in absolute terms, contained the more actionable signals - particularly where multiple insiders at the same company bought within a short window.

References

Lakonishok, J., & Lee, I. (2001). Are insider trades informative? *The Review of Financial Studies*, 14(1), 79–111. <https://doi.org/10.1093/rfs/14.1.79>

U.S. Securities and Exchange Commission. (2025). *Insider transactions data sets*. Retrieved February 2026, from <https://www.sec.gov/data-research/sec-markets-data/insider-transactions-data-sets>